IMPERIAL

A Musical Application of Cluster Analysis

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Introduction

- Cluster analysis is a technique where observations are separated into 'clusters' that partition the set of observations.
- In this project, multiple clustering algorithms are applied and evaluated to find an optimal clustering of a music data set.
- There is a stigma surrounding Taylor Swift's music. This project aims to show that there exists variety within her style.
- The code used can be found at https://github.com/lewistang-uk/A-Musical-Application-of-Cluster-Analysis/ - the clustering and visualisation was done in Python using numpy, pandas, scikit-learn, matplotlib and seaborn.

Dataset

The data used is a subset of Swift's works to date, obtained from Kaggle (Priester, 2024). [1]

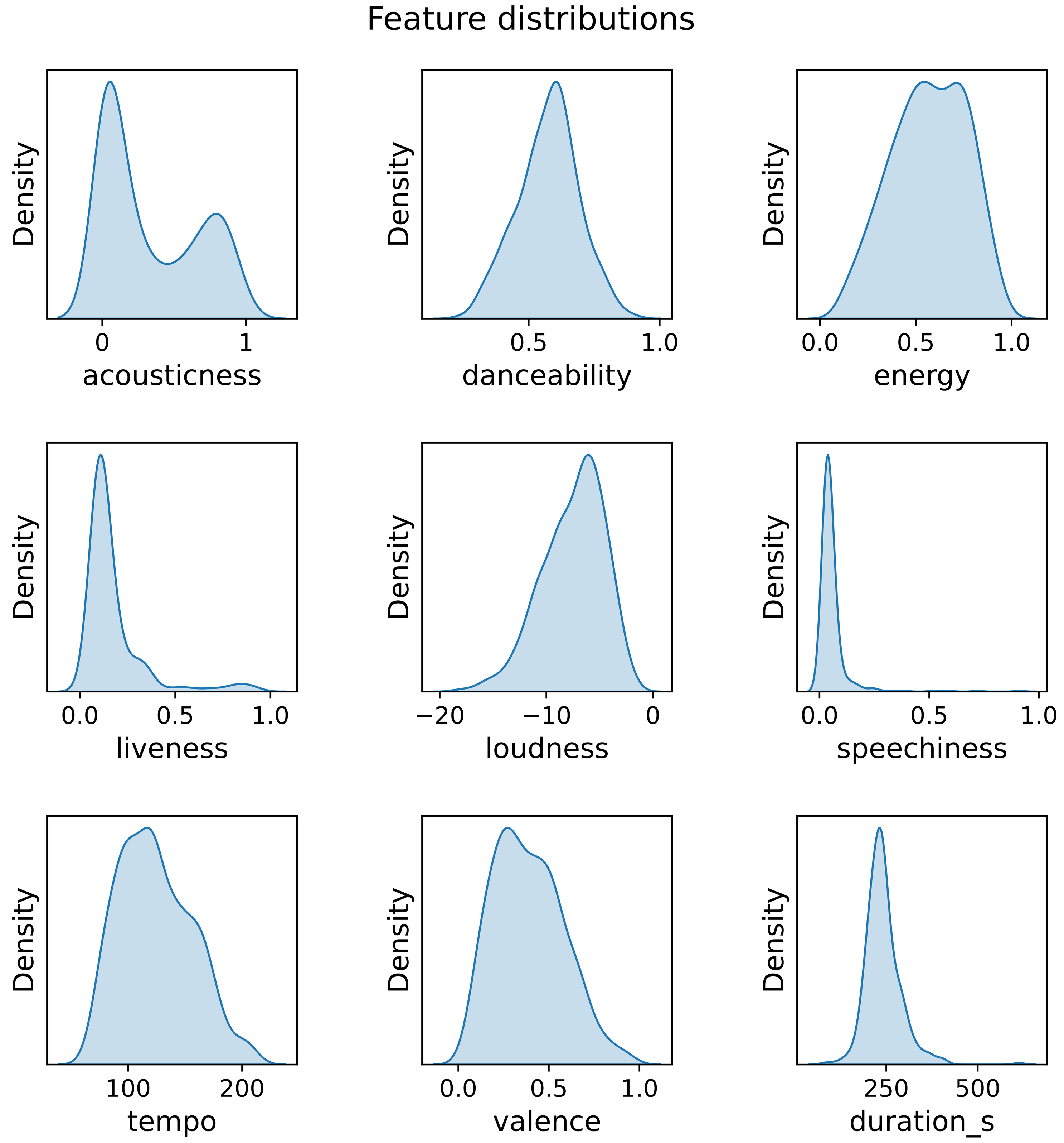

Figure 1: Density estimate plots of musical features.

Clustering algorithms

- K-Means Clustering (MacQueen, 1967) [2]
  - Assign each point to a cluster by its nearest centroid.
  - Adjust centroid position to minimise variance within clusters.
  - Repeat until convergence.
- Agglomerative Clustering (Ward, 1963) [3]
  - Merge the most similar pair of clusters recursively.
  - The resulting hierarchical structure implies a clustering.

Determining number of clusters

- Elbow (Thorndike, 1953) [4]
  - Plot WCSS (Within-Cluster Sum of Squares) against clusters.
  - Find the point where the gradient drops sharply.
- Silhouette (Rousseeuw, 1987) [5]
  - Measure the similarity of a point to its own cluster compared to other clusters.
  - Find the average of this metric and maximise by varying the number of clusters.

Application of clustering

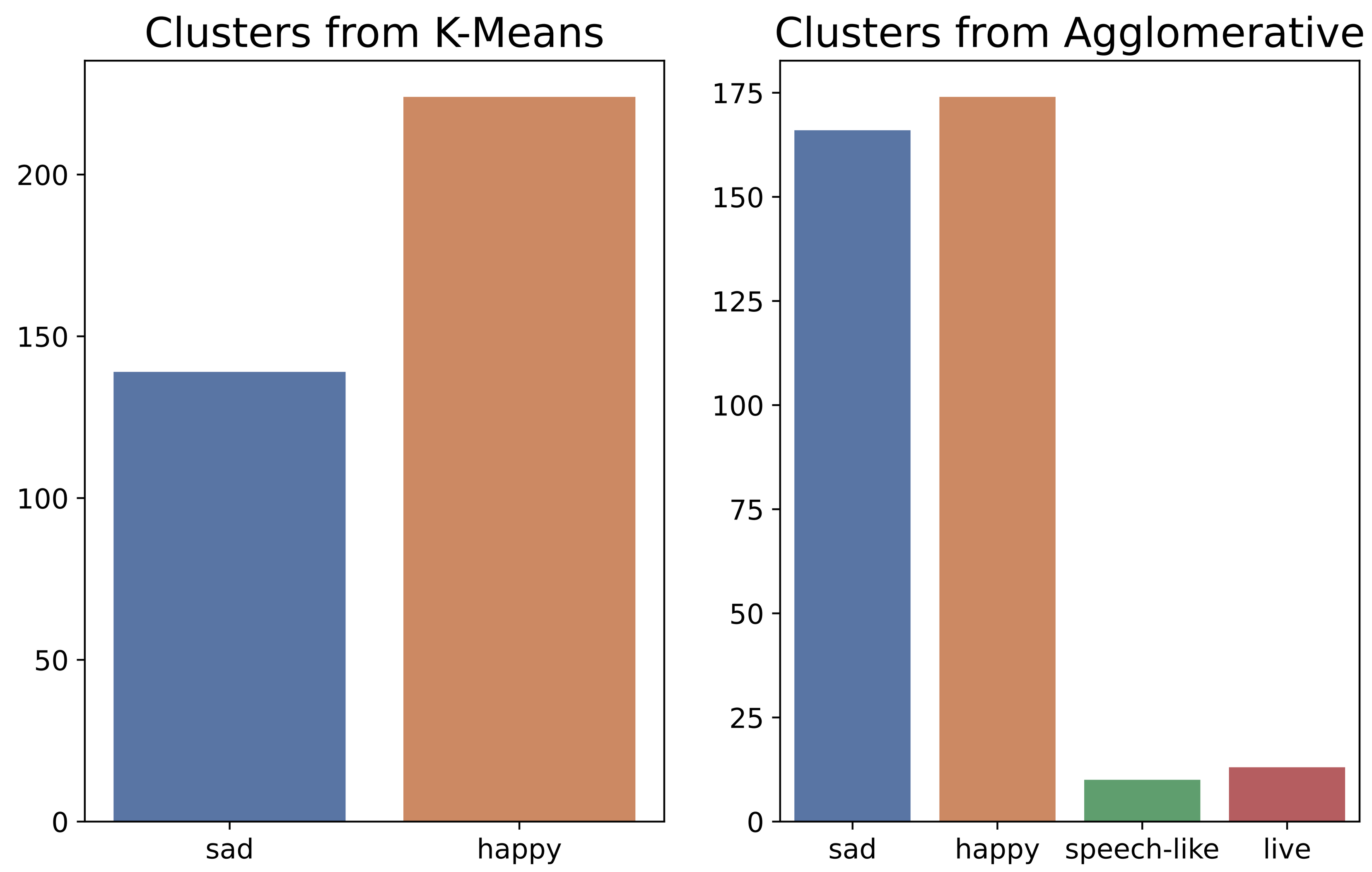

Figure 2: Number of observations in each cluster after feature selection for both clustering algorithms.

Examples of differing cluster labels

song	K-Means	Agglomerative
Question...?	happy	speech-like
closure	sad	speech-like
Better Than Revenge - Live/2011	happy	live
Ours - Live/2011	sad	live
False God	sad	speech-like
Bejeweled	happy	sad
All Too Well	happy	sad

Summary

- Agglomerative finds more patterns than K-Means — K-Means assumes that clusters are roughly equal in cardinality (Bishop, 2006). [6]
- Both clustering algorithms point to two main clusters, showing that Swift's music may not all be in the same style.
- Neither set of clustering labels is perfect - 'Bejeweled' should be a 'happy' song and 'All Too Well' should be a 'sad' song.
- In an interview, Swift (2022) [7] described three different styles of lyric writing. Analysis of lyrics could point to a different clustering.

References

[1] Priester, J (2024), Taylor Swift Spotify Dataset, v. 21. Retrieved 24/05/26 from https://www.kaggle.com/datasets/jarredpriester/taylor-swift-spotify-dataset/ [2] MacQueen, J. (1967) Some methods for classification and analysis of multivariate observations. Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability. 1, 281–297. [3] Ward, J.H. (1963) Hierarchical grouping to optimize an objective function. Journal of the American Statistical Association. 58 (301), 236–244. [4] Thorndike, R.L. (1953) Who belongs in the family?. Psychometrika. 18(4), 267–276. [5] Rousseeuw, P.J. (1987) Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics. 20(1), 53–65. [6] Bishop, C.M. (2006) Pattern Recognition and Machine Learning. New York, Springer. [7] Rogerson, B. (2022) Taylor Swift reveals the 3 lyric 'genre categories' she uses when she's writing songs. MusicRadar, 21 September. Retrieved 08/06/26 from https://www.musicradar.com/news/taylor-swift-songwriting-secret